Nat x Nat is enumerable via Cantor's zig-zag enumeration

The Open Logic Project contributors

openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-sets-functions-relations-size-of-sets/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the span's !!{enumerable} shorthand renders literally in the shipped minimal preamble (intake's 11:57 no-shim state) - recorded here rather than silently normalized. The zig-zag exhibit itself sits in the preceding explain block and is pinned by position, not by label.

Editorial note. The label natsquaredenumerable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Cantor's zig-zag array and its enumeration order. Same section, the explain block immediately preceding the proposition (source lines ~2156-2184): the doubly indexed array of Nat x Nat displayed row by row, named 'Cantor's zig-zag method', with the enumeration order written out explicitly as the tuple sequence (0,0), (0,1), (1,0), (0,2), (1,1), (2,0), ... Lines 2191-2193: the map f is defined as sending each k to the tuple at the k-th position of exactly this sequence - the whole proof lives on the displayed exhibit.

Proposition 0.1. $\mathbb{N} \times \mathbb{N}$ *is enumerable.*

Proof. Let $f\colon \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ take each $k \in \mathbb{N}$ to the tuple $\langle n, m \rangle \in \mathbb{N} \times \mathbb{N}$ such that k is the value of the nth row and mth column in Cantor's zig-zag array. □